

One center fit serves every K; only step 4 is repeated per K

1. Features

Pre-exam record:
grades 10, 11 and
12 (term I), school,
province, area
(F_{dt-cn} , 160 cols)

2. Center $\hat{y}$

XGBoost, squared loss
bag of $B^* = 20$
(or one RMSE tuning;
no tail fitness)

3. OOF pairs

5-fold out-of-fold
($\hat{y}_{oof}, y$) on the
training set; refit on
all training rows

4. Decision layer

for each K: fit g_K on
($\hat{y}_{oof}, y$) with
 $w_K(y)$: weighted
isotonic (R2) or
residual bins (R1)

5. Outputs

$g_K(\hat{y})$: decision
score under cost K
 $P(Y \geq c \mid x)$: for
fixed thresholds
(not shown to students)

HPO with a tail-weighted fitness
and weighted training R8 act here

Output stretch R5 and the other
layer estimators (R3-R7) act here